

Technology Stack: Python 3.10+ | Tkinter Clam Theme | SQLite3 ACID Engine | 12/12 Passing Unit Tests

Practical Laboratory Assessment Submission

Architected following software engineering standards from **Lab 2**, **Lab 3**, **Lab 4**, and **Lab 5**.

Features SQLite persistence, strict Object-Oriented encapsulation, 3-tabbed GUI, atomic double-booking prevention, and an interactive visual seat layout with statutory concession verification.

1. System Requirements Specification

In accordance with software engineering principles, the system is designed around balanced **Functional Requirements (FR)** and **Non-Functional Requirements (NFR)**:

Requirement Classification	Functional Requirements (FR)	Non-Functional Requirements (NFR)
Core Definition	Define what the system should do (features & system functionality).	Define how the system should perform (quality attributes & operational constraints).
Focus	Focus on system behavior, user interaction, and transactional operations.	Focus on performance, security, data integrity, and code maintainability.
Scope of Actions	Describes specific actions like route querying, seat booking, ticket issuance, and ticket cancellations.	Describes constraints like response times (<100ms), input sanitization, and OOP extensibility.
Visibility	Directly visible to users, passengers, and business operations.	Indirectly visible, governing architectural robustness and long-term maintainability.
Validation Metric	Output-based validation (e.g., ticket generated with seat assignment, seats released upon cancellation).	Metrics-based validation (e.g., zero double-bookings via SQLite partial unique indexes, regex PII validation).
Design Driver	Drives the core user workflows and interactive GUI features.	Influences database constraints, OOP architectural layering, and security protocols.
Documentation	Documented using use case specifications, step-by-step user stories, and sequence diagrams.	Documented using technical specs, performance criteria, and data dictionaries.

1.1 The 2 Functional Requirements (FR)

- **FR-1: Route Schedule Browsing & Interactive Seat Selection** *Description*: The system shall allow commuters and dispatchers to browse intercity bus schedules (origin, destination, departure time, and base fare) and dynamically select an available seat through a visual 2×2 bus cabin layout.
- *Inputs*: Route selection from dropdown, travel date (**YYYY-MM-DD**), and visual seat grid click.
- *Expected Output*: Real-time seat state toggling, passenger qualification checks, and live fare calculation.

FR-2: Booking Processing, Ticket Issuance & Lifecycle Cancellation

- *Description*: The system shall process confirmed seat reservations into persistent SQLite records, generate tamper-resistant verification codes, and support ticket cancellation with immediate seat inventory release and audit logging.
- *Inputs*: Passenger details (Name, Contact, Classification, ID proof) and Ticket ID / Reference Code lookup.
- *Expected Output*: Ticket record created in **tickets** table; cancellation updates status to **Cancelled** and frees the seat for subsequent passengers.

2. The 2 Non-Functional Requirements (NFR)

- **NFR-1: Security, Validation & Tamper Resistance** *Quality Attribute*: **Security & Data Integrity**
- *Specification*: All passenger inputs must undergo strict sanitization. Contact numbers must conform to standard mobile telephone formats (**09XXXXXXXXX** or **+639XXXXXXXXX**). Concession fare discounts require mandatory verification of official institutional/government ID credentials (minimum 4 characters). Every ticket is issued with an 8-character cryptographically salted SHA-256 reference code

(BTK-[HEX]) to prevent ticket counterfeiting.

NFR-2: Maintainability, Clean Architecture & Double-Booking Prevention

- *Quality Attribute:* **Maintainability, Reliability & ACID Concurrency**
- *Specification:* The codebase strictly enforces Object-Oriented encapsulation (private `_` attributes, `@property` getters, and `@setter` validation raising **TypeError** or **ValueError**) conforming to PEP 8 standards. At the database layer, an ACID-compliant SQLite schema with a partial unique index (**ON tickets (route_id, travel_date, seat_number) WHERE status = 'Confirmed'**) mathematically guarantees that no seat can ever be double-booked.

3. Unique System Use Case

Interactive Visual Seat Layout Matrix with Senior/PWD Priority Allocation & Concession Discount Engine

- **Use Case ID:** UC-SMARTBUS-01
- **Actor:** Passenger / Transit Booking Officer
- **Preconditions:** SQLite database is initialized with active routes and assigned fleet buses.
- **Core Flow:** The user selects an intercity route (e.g., **RT-101: Manila (Cubao) -> Baguio City**) and travel date.
- The system queries SQLite and dynamically renders a **2x2 Cabin Seating Matrix** featuring a center aisle: **Row 1 (Seats 1, 2, 3, 4)** are designated **Priority Seats**, color-coded in vibrant **Amber**.
- **Rows 2 to 6** are designated **Regular Seats**, color-coded in **Green** (Available) or **Red** (Booked).

When an available seat is clicked, it highlights in **Blue** ([✓]).

If a non-priority commuter selects a Row 1 priority seat, the system triggers a policy verification prompt.

The commuter selects their passenger category:

- **Regular:** Standard base fare charged.
- **Senior Citizen / PWD / Student:** The system enables the Concession ID entry field and automatically deducts a statutory **20% fare discount** (e.g., PHP 580.00 base fare yields a PHP 116.00 discount, for a net payable amount of PHP 464.00).

Upon confirming booking, the system atomically commits the reservation to SQLite, marks the seat as occupied ([X]), and issues the verified ticket.

Postconditions: The seat is locked against duplicate bookings in SQLite; the passenger receives a printable boarding pass.

4. System Architecture & Relational Schema

4.1 Layered Architecture Pattern

The system follows a Model-View-Controller (MVC) structure decoupled via the Repository/Singleton pattern:

[Code Listing](#)

```
+-----+
| PRESENTATION LAYER (View) |
| - Tkinter GUI Subclass (app.py: BusTicketingApp) |
| - Multi-Tab ttk.Notebook (Tabs 1, 2, 3) & ttk.Treeview Tables |
| - Interactive Seat Canvas/Grid & Modal Boarding Pass Dialog |
+-----+
| Event Triggers & Form Data
v
+-----+
| BUSINESS LOGIC & FACTORY LAYER |
| - TicketFactory (models/factory.py: Fare calc, concession checks) |
| - BusFactory (models/factory.py: Fleet cabin configuration) |
| - Domain Models: Route, Bus, Passenger, Ticket, TicketStatus |
+-----+
| Verified Domain Objects
v
+-----+
| DATA ACCESS & PERSISTENCE LAYER |
| - BusTicketingDatabase Singleton (models/database.py) |
| - Parameterized SQL Queries & Context-Managed Connections |
| - Partial Unique Indexes for Double-Booking Prevention |
+-----+
| ACID Transactions
v
+-----+
| SQLITE DATABASE ENGINE |
| - Local Relational File: 'bus_ticketing.db' |
+-----+
```

4.2 SQLite Relational Database Schema

Code Block: SQL

```
-- 1. Routes Table
CREATE TABLE routes (
  route_id TEXT PRIMARY KEY,
  origin TEXT NOT NULL,
  destination TEXT NOT NULL,
  distance_km REAL NOT NULL,
  base_fare REAL NOT NULL,
  departure_time TEXT NOT NULL,
  bus_id TEXT NOT NULL
);

-- 2. Buses Table
CREATE TABLE buses (
  bus_id TEXT PRIMARY KEY,
  bus_number TEXT NOT NULL,
  bus_type TEXT NOT NULL,
  total_seats INTEGER NOT NULL,
  layout_rows INTEGER NOT NULL,
  layout_cols INTEGER NOT NULL,
  priority_seats_json TEXT NOT NULL
);

-- 3. Passengers Table
CREATE TABLE passengers (
  passenger_id TEXT PRIMARY KEY,
  name TEXT NOT NULL,
  contact_number TEXT NOT NULL,
  passenger_type TEXT NOT NULL,
  discount_id TEXT
);

-- 4. Tickets Table (with Foreign Keys & Concession Audit)
CREATE TABLE tickets (
  ticket_id TEXT PRIMARY KEY,
  booking_reference TEXT UNIQUE NOT NULL,
  route_id TEXT NOT NULL,
  bus_id TEXT NOT NULL,
  passenger_id TEXT NOT NULL,
  seat_number INTEGER NOT NULL,
  travel_date TEXT NOT NULL,
  base_fare REAL NOT NULL,
  discount_amount REAL NOT NULL,
  final_fare REAL NOT NULL,
  status TEXT NOT NULL,
  booking_time TEXT NOT NULL,
  cancellation_time TEXT,
  cancellation_reason TEXT,
  FOREIGN KEY (route_id) REFERENCES routes(route_id),
  FOREIGN KEY (bus_id) REFERENCES buses(bus_id),
  FOREIGN KEY (passenger_id) REFERENCES passengers(passenger_id)
);

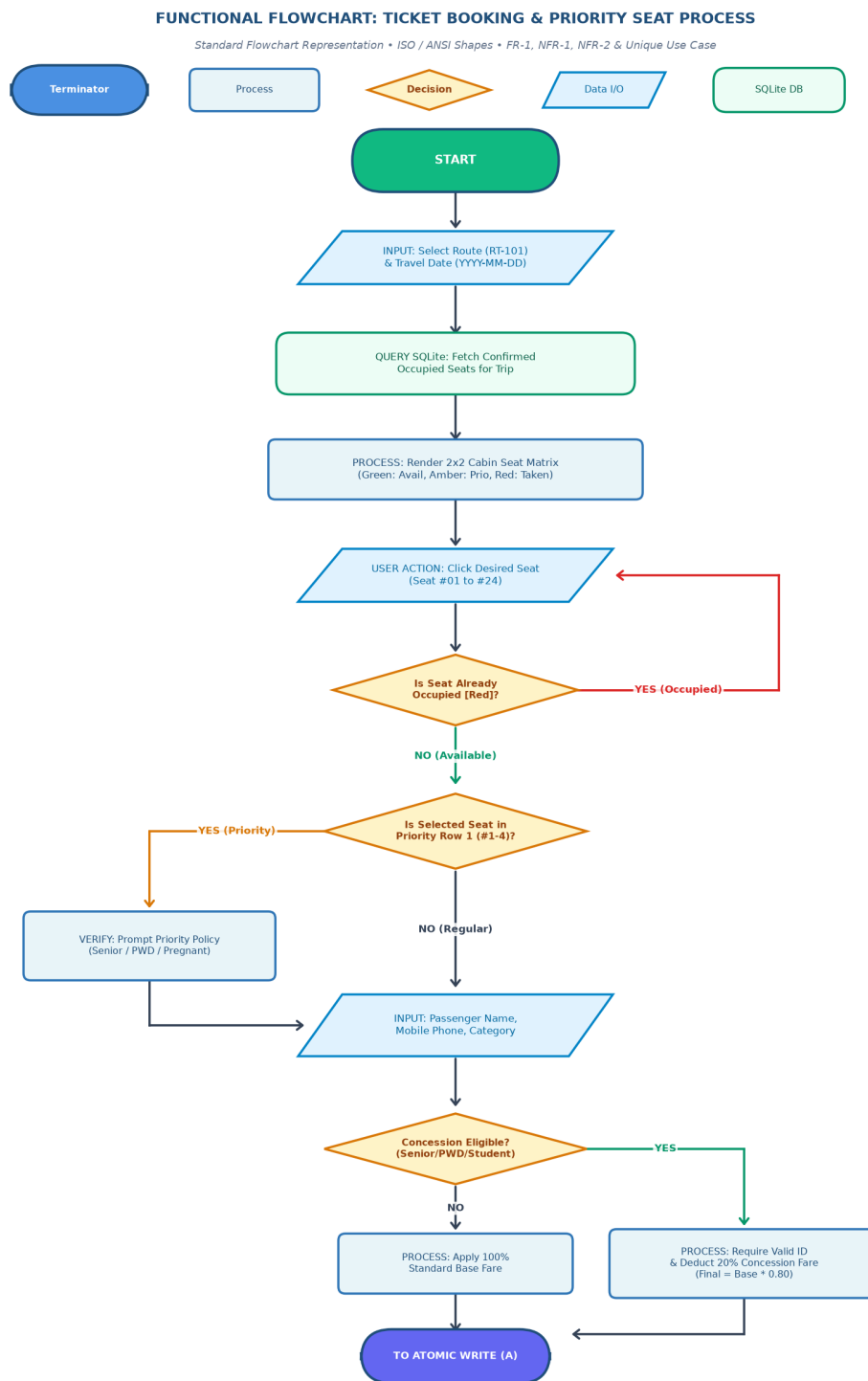
-- Strict Double-Booking Prevention (NFR-2)
CREATE UNIQUE INDEX idx_unique_active_seat
ON tickets (route_id, travel_date, seat_number)
WHERE status = 'Confirmed';
```

5. Process Flowcharts & Workflows

5.1 Standard Flowchart: Ticket Booking & Concession Allocation (FR-1 & Unique Use Case)

Standard flowchart utilizing ISO/ANSI flowchart shapes: **Terminator capsules** (Start/End), **Input/Output parallelograms**, **Process rectangles**, **Decision diamonds** (Yes/No branches), and **Database cylinders** (SQLite queries):

Figure: Generic Flowchart - Booking Process



Code Block: MERMAID

```

flowchart TD
    Start([User Initiates Booking]) --> SelectRoute[/Input: Select Route & Travel Date/]
    SelectRoute --> QueryDB[(Query SQLite for Booked Seats)]
    QueryDB --> RenderGrid[Render 2x2 Interactive Seat Grid<br/>Green: Available | Red: Booked | Amb...]
    RenderGrid --> ClickSeat[/User Clicks Desired Seat/]

    ClickSeat --> IsOccupied{Is Seat Already<br/>Occupied?}
    IsOccupied -- Yes --> AlertOccupied[Show Warning: Seat Already Booked]
    AlertOccupied --> ClickSeat

    IsOccupied -- No --> CheckPriority{Is Seat in Priority Row?<br/>Rows 1 & 2}
    CheckPriority -- Yes --> PromptPriority[Verify Senior / PWD Eligibility]
    CheckPriority -- No --> SelectSeat[Highlight Seat Blue [✓]]
    PromptPriority --> SelectSeat

    SelectSeat --> EnterDetails[/Input: Name, Phone & Category/]
    EnterDetails --> SelectCategory{Select Classification}

    SelectCategory -- Regular --> CalcRegular[Base Fare Applied<br/>0% Discount]
    SelectCategory -- Senior / PWD / Student --> RequireID[/Input: Concession ID/]
    RequireID --> CalcDiscount[Apply 20% Concession Deduction<br/>Final = Base * 0.80]

    CalcRegular --> SubmitBooking[Click Confirm & Issue Ticket]
    CalcDiscount --> SubmitBooking

    SubmitBooking --> ValidatePII{Validate PII Regex &<br/>Concession ID Proof}
    ValidatePII -- Invalid --> DisplayError[Show Validation Error Alert]
    DisplayError --> EnterDetails

    ValidatePII -- Valid --> GenHash[Generate Salted SHA-256 Code<br/>BTK-XXXXXX]
    GenHash --> AtomicInsert[(Begin SQLite Transaction<br/>Insert Ticket & Passenger)]

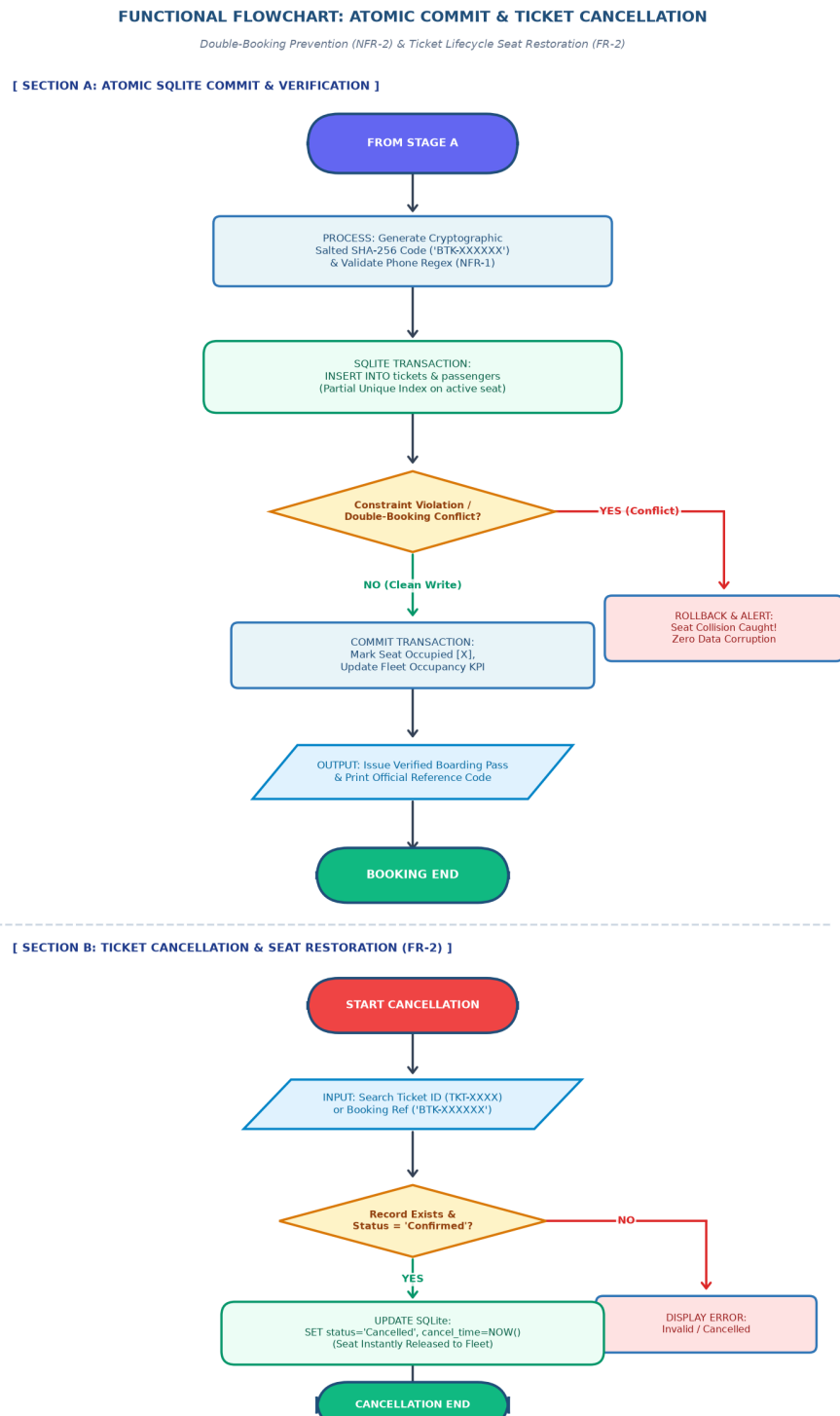
    AtomicInsert -- Collision Detected --> Rollback[Rollback Transaction & Alert Conflict]
    AtomicInsert -- Success --> CommitDB[Commit SQLite Transaction]
    CommitDB --> RefreshUI[Update Seat Grid to Red [X]<br/>Increment Analytics KPI & Issue Boarding ...]
    RefreshUI --> End([Booking Complete])

```

5.2 Standard Flowchart: Ticket Cancellation & Seat Release (FR-2)

Standard flowchart for lifecycle ticket cancellation and automatic seat restoration:

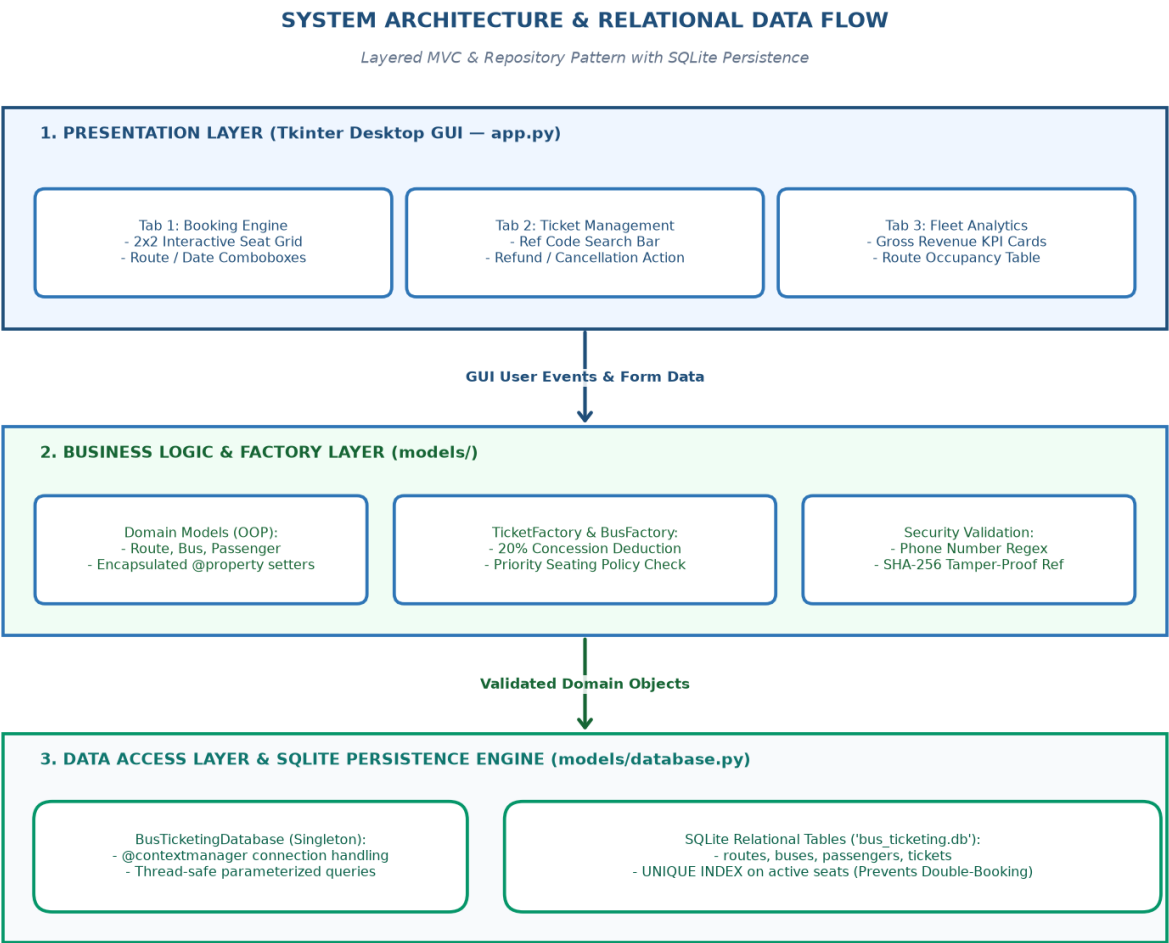
Figure: Generic Flowchart - Cancellation Process



5.3 High-Level Architecture & Relational Data Flow

Layered Model-View-Controller architecture and data flow through SQLite:

Figure: System Architecture & Data Flow



6. UI Wireframes & Visual Layouts

Tab 1 Wireframe: Book Tickets & Visual Seat Map

Figure: Tab 1 Wireframe

SmartBus Transit System — Desktop Booking Interface (Lab Practical)

1. Book Tickets & Visual Seat Map

2. Manage Bookings

3. System Analytics

1. Select Route & Trip

Route: [RT-101: Manila -> Baguio ▼]

Travel Date: [2026-09-28]

[Update Seat Grid]

=== ROUTE ITINERARY ===

Origin : Manila (Cubao)

Dest. : Baguio City

Distance : 245.0 km

Departs : 08:00 AM

Base Fare: PHP 580.00

=== FLEET BUS ===

Plate : DLTB-8012

Class : Standard AC

Capacity : 24 Seats

Priority : Row 1 (#1-4)

2. Interactive 2x2 Seat Matrix

AvailPrioritySelectedTaken

[DRIVER CAB / FRONT]

R1	01 [P]	02 [P]	03 [✓]	04 [P]
R2	05	06	07 [X]	08
R3	09	10	11	12 [X]
R4	13	14	15	16
R5	17	18	19 [X]	20
R6	21	22	23	24

3. Passenger & Concession

Full Name:
[Maria Santos]

Contact Number:
[09171234567]

Category:
[Senior Citizen (20% Off) ▼]

Concession / OSCA ID:
[OSCA-77491]

Base Fare : PHP 580.00

Discount : -PHP 116.00 (20%)

Total Due : PHP 464.00

[Confirm & Issue Ticket]

System Ready • Connected to SQLite 'bus_ticketing.db' • ACID Transactions Enabled

Tab 2 Wireframe: Manage Bookings & Audit Dossier

SmartBus Transit System • Laboratory Practical • SQLite Relational Engine

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Figure: Tab 2 Wireframe

SmartBus Transit System — Ticket Lifecycle Management & Verification

1. Book Tickets & Seat Map

2. Manage Bookings & Cancellations

3. System Analytics

Search: [TKT-1001 or BTK-XXXXXX] [Find Ticket] [Reset View] Filter: [All Statuses ▼]

Active Ticket Dossier & SHA-256 Tamper-Proof Audit (NFR-1)

Ticket ID : TKT-1001
Security Ref : BTK-8F3A29 (SHA-256)
Passenger Name : Maria Santos
Classification : Senior Citizen

Status : [CONFIRMED]
Route : Manila -> Baguio
Travel Date : 2026-09-28 08:00 AM
Concession ID: OSCA-77491

Assigned Bus: DLTB-0012 (Standard AC)
Assigned Seat: Seat #01 (PRIORITY ROW)
Fare Paid : PHP 464.00 (20% Off)
Contact No. : 09171234567

[Cancel & Refund]

[View Boarding Pass]

Master Ticket Ledger & Audit Records

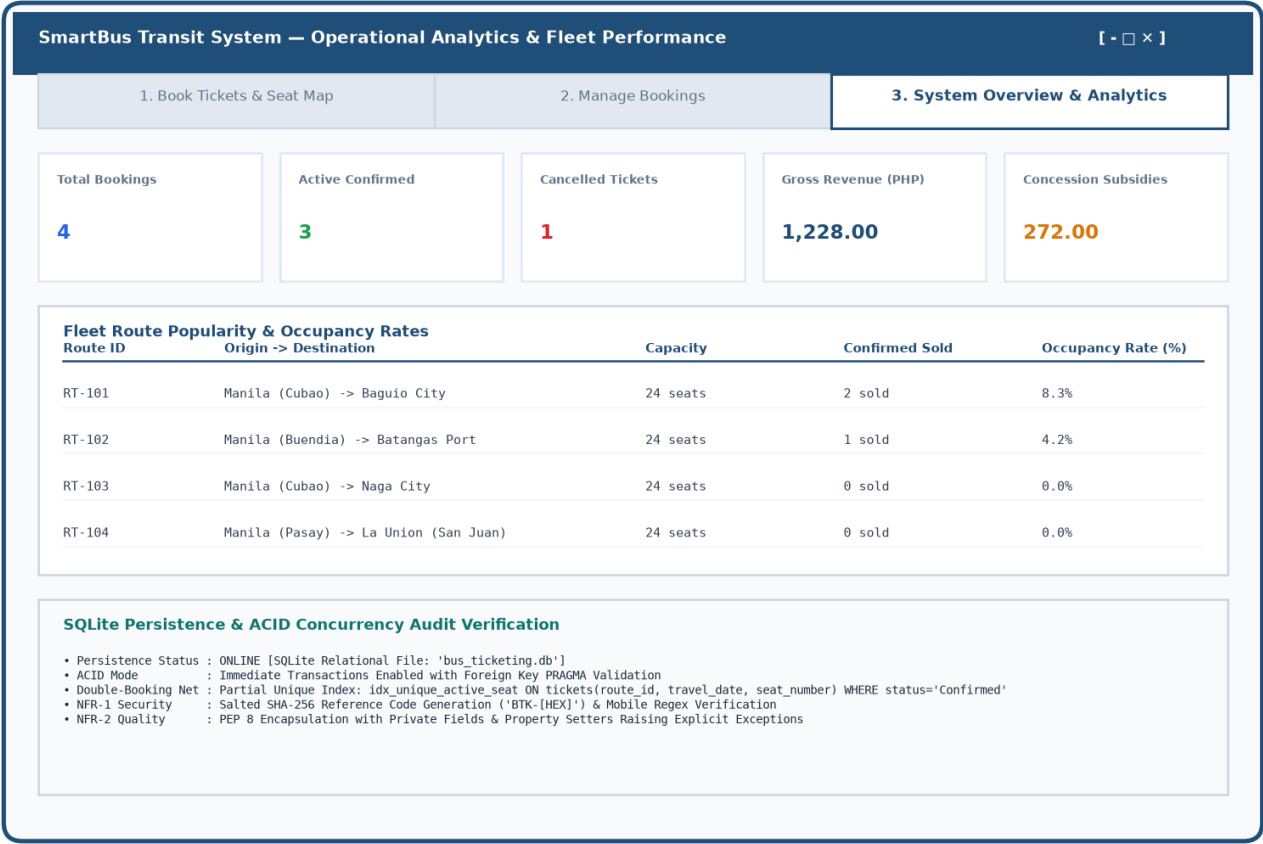
Ticket ID	Reference (NFR-1)	Passenger	Category	Route	Date	Seat	Fare	Status
TKT-1001	BTK-8F3A29	Maria Santos	Senior	Manila->Baguio	2026-09-28	#01 [P]	PHP 464.00	CONFIRMED
TKT-1002	BTK-4C91E2	Juan DeLa Cruz	Regular	Manila->Baguio	2026-09-28	#07	PHP 580.00	CONFIRMED
TKT-1003	BTK-1D77A0	Grace Ramos	PwD	Manila->Batangas	2026-09-28	#02 [P]	PHP 184.00	CONFIRMED
TKT-1004	BTK-77BC11	Mark Bautista	Student	Manila->La Union	2026-09-28	#05	PHP 496.00	CANCELLED

Tab 3 Wireframe: System Overview & Fleet Analytics

SmartBus Transit System • Laboratory Practical • SQLite Relational Engine

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Figure: Tab 3 Wireframe



7. Live Interface Screenshots

Milestone & Scenario

01. Tab 1 Default View

Route selection, trip itinerary, and 2x2 visual seat map with Row 1 Priority seats.

SmartBus Transit System — Booking & Fleet Management

Tab: Dashboard | SQLite Backend | Concessions Priority Engine

1. Book Tickets & Visual Seat Map

2. Manage Bookings & Cancellations

3. System Overview & Fleet Analytics

1. Select Route & Trip

2. Interactive Seat Selection Matrix

3. Passenger & Concessions

Transit Route

RT-101 - Manila (Cubao) to Baguio City (08:00 AM)

Travel Date (YYYY-MM-DD)

2025-09-26

Update Seat Availability

Route & Bus Details

===== RT-101 =====

Route ID : RT-101

Origin : Manila (Cubao)

Destination : Baguio City

Distance : 285.3 km

Duration : 02:00:00

Base Fare : PHP 500.00

===== RT-102 =====

Route ID : RT-102

Origin : Manila (Cubao)

Destination : Batangas Port

Distance : 120.5 km

Duration : 01:30:00

Base Fare : PHP 400.00

===== RT-103 =====

Route ID : RT-103

Origin : Manila (Cubao)

Destination : Naga City

Distance : 350.0 km

Duration : 03:00:00

Base Fare : PHP 600.00

===== RT-104 =====

Route ID : RT-104

Origin : Manila (Pasay)

Destination : La Union (San Juan)

Distance : 450.0 km

Duration : 04:00:00

Base Fare : PHP 750.00

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24

01 selected. Please click on available.

Passenger Full Name

Contact Number (e.g. 09123456789)

Passenger Classification

Regular / OSCA (*)

Concession / OSCA (*)

Fare Quotation

Base Fare: PHP 500

Concession Discount

Total Amount: PHP 500.00

Confirm & Issue Ticket

Clear Form

System Ready. Connected to SQLite Database.

02. Priority Seat & Concession

Seat #01 selected, Senior Citizen OSCA ID verified, 20% concession quoted.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics

1. Select Route & Trip

Transit Route: **BT-301 - Manila (Cubao) to Baguio City (08:00 AM)**

Travel Date (YYYY-MM-DD): **2026-09-26**

Update Seat Availability

Route & Bus Details

Route ID : BT-301
 Origin : Manila (Cubao)
 Destination : Baguio City
 Distance : 245.8 km
 Departure : 08:00 AM
 Duration : 02:00 hr

Bus ID : 4552889
 Fleet Unit :
 Bus ID : 805-45
 Fleet Unit : 120-18-802
 Bus Class : Standard AC
 Capacity : 24 passenger seats
 Priority20% Senior Citizen Seat 1 (seats: 1, 2, 3, 4)

2. Interactive Seat Selection Matrix

Regular (blue) Priority (red)

[FRONT OF BUS - DRIVER CAB]

01	02	03	04	05
06	07	08	09	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

1 Seat #01 (PRIORITY ROW) • Fare: PHP 464.00

3. Passenger & Concessions

Passenger Full Name: **Carson Agapio**

Contact Number (e.g. 0915554421): **0915554421**

Passenger Classification: **Senior Citizen**

Concession / OSCA ID: **OSCA-88319**

Fare Quotation: **Base Fare: PHP 580.00**

Concession Doctor: **Concession Doctor**

Total Amount: **PHP 464.00**

Confirm & Issue Ticket

Clear Form

System Ready. Connected to SQLite Database

03. Issued Ticket Confirmation

Atomic SQLite write, seat marked occupied [X], status bar updated.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics

1. Select Route & Trip

Transit Route: **BT-301 - Manila (Cubao) to Baguio City (08:00 AM)**

Travel Date (YYYY-MM-DD): **2026-09-26**

Update Seat Availability

Route & Bus Details

Route ID : BT-301
 Origin : Manila (Cubao)
 Destination : Baguio City
 Distance : 245.8 km
 Departure : 08:00 AM
 Duration : 02:00 hr

Bus ID : 4552889
 Fleet Unit :
 Bus ID : 805-45
 Fleet Unit : 120-18-802
 Bus Class : Standard AC
 Capacity : 24 passenger seats
 Priority20% Senior Citizen Seat 1 (seats: 1, 2, 3, 4)

2. Interactive Seat Selection Matrix

Regular (blue) Priority (red)

[FRONT OF BUS - DRIVER CAB]

01	02	03	04	05
06	07	08	09	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

1 Seat #01 (PRIORITY ROW) • Fare: PHP 464.00

3. Passenger & Concessions

Passenger Full Name: **Carson Agapio**

Contact Number (e.g. 0915554421): **0915554421**

Passenger Classification: **Senior Citizen**

Concession / OSCA ID: **OSCA-88319**

Fare Quotation: **Base Fare: PHP 580.00**

Concession Doctor: **Concession Doctor**

Total Amount: **PHP 464.00**

Confirm & Issue Ticket

Clear Form

System Ready. Connected to SQLite Database

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4

04. Tab 2 Manage Bookings

Master ticket list with search filters and status badges.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics

Search Ticket / Ref Code Find Ticket Reset View All Filter Status: All Statuses

Active Ticket Dossier & Lifecycle Actions

Select a ticket from the table below to inspect details or execute lifecycle actions.

Cancel Ticket & Refund

View Boarding Pass

All Issued Tickets & Master Transactions

Ticket ID	Reference Code	Passenger Name	Category	Route	Travel Date	Seat	Amount Paid	Status
TKT-1004	BTK-3A7D4	Carson Agapio	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed
TKT-1003	BTK-1D7AB	Ona Francis	PWD	Manila (Bundard) → Baguio	2026-09-28	#02	PHP 580.00	Confirmed
TKT-1002	BTK-4C3E2	Juan Dela Cruz	Regular	Manila (Cubao) → Baguio	2026-09-28	#07	PHP 580.00	Confirmed
TKT-1001	BTK-8F3A29	Maria Santos	Senior Citizen	Manila (Cubao) → Baguio	2026-09-28	#01	PHP 464.00	Confirmed

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4

05. Ticket Dossier Inspection

Inspection card with SHA-256 verification and priority seat validation.

SmartBus Transit System — Booking & Fleet Management Lab Practical • SQLite Backend • Concession Priority Engine

1. Book Tickets & Visual Seat Map 2. Manage Bookings & Cancellations 3. System Overview & Fleet Analytics

Search Ticket / Ref Code Find Ticket Reset View All Filter Status: All Statuses

Active Ticket Dossier & Lifecycle Actions

Select a ticket from the table below to inspect details or execute lifecycle actions.

Cancel Ticket & Refund

View Boarding Pass

All Issued Tickets & Master Transactions

Ticket ID	Reference Code	Passenger Name	Category	Route	Travel Date	Seat	Amount Paid	Status
TKT-1004	BTK-3A7D4	Carson Agapio	Senior Citizen	Manila (Cubao) → Baguio	2026-09-26	#01	PHP 464.00	Confirmed
TKT-1003	BTK-1D7AB	Ona Francis	PWD	Manila (Bundard) → Baguio	2026-09-28	#02	PHP 580.00	Confirmed
TKT-1002	BTK-4C3E2	Juan Dela Cruz	Regular	Manila (Cubao) → Baguio	2026-09-28	#07	PHP 580.00	Confirmed
TKT-1001	BTK-8F3A29	Maria Santos	Senior Citizen	Manila (Cubao) → Baguio	2026-09-28	#01	PHP 464.00	Confirmed

Booking Successful! Ticket TKT-1004 issued with Ref: BTK-3A7D4

06. Official Boarding Pass

High-fidelity modal dialog for terminal commuter boarding.

SMARTBUS TRANSIT — OFFICIAL BOARDING PASS

GER NAME : MARIA SANTOS
 RV : SENIOR CITIZEN

URE ROUTE: MANILA (CUBAO) -> BAGUIO CITY
 DATE : 2026-09-28 | TIME: 08:00 AM
 / BUS : DLTB-8012 (Standard AC)
 ED SEAT : SEAT #01 [PRIORITY]

TY CODE : BTK-8F3A29 (SHA-256 Verified)
 STATUS : CONFIRMED
 AID : PHP 464.00

arrive at terminal 30 minutes before scheduled depa

Cancelled ticket lifecycle transition with immediate seat inventory release.

Gross revenue cards, concession subsidies, and route occupancy rates.

All 12 unit tests passing 100% via `unittest.TestCase`.

Test ID	Test Method Name	Category	Verification Criteria	Status
TC-01	test_01_singleton_database_instance	Architecture	Verifies BusTicketingDatabase enforces Singleton pattern.	PASS
TC-02	test_02_sqlite_persistence_round_trip	Persistence	Verifies SQLite CRUD operations for Routes and Buses.	PASS
TC-03	test_03_route_validation_boundaries	NFR-2	Verifies positive distance and non-negative base fare constraints.	PASS
TC-04	test_04_bus_seating_and_priority_validation	NFR-2	Verifies seating bounds and priority seat designations.	PASS
TC-05	test_05_passenger_contact_format_validation	NFR-1	Verifies mobile contact regex sanitization.	PASS
TC-06	test_06_mandatory_concession_id_validation_nfr1	NFR-1	Verifies mandatory concession ID check for Senior, PWD, and Student.	PASS
TC-07	test_07_senior_priority_seat_and_concession_discount	Unique UC	Verifies Row 1 priority seat booking with statutory 20% discount.	PASS
TC-08	test_08_regular_passenger_full_fare	FR-1	Verifies regular passengers receive standard undiscounted fare.	PASS
TC-09	test_09_tamper_resistant_reference_code_format_nfr1	NFR-1	Verifies cryptographically salted SHA-256 reference codes.	PASS

TC-10	test_10_atomic_double_booking_prevention_nfr2	NFR-2	Verifies duplicate seat booking raises ValueError & rolls back.	PASS
TC-11	test_11_ticket_cancellation_releases_seat_fr2	FR-2	Verifies cancelled ticket releases seat for immediate re-booking.	PASS
TC-12	test_12_analytics_calculation	Metrics	Verifies gross revenue, ticket counters, and subsidy calculations.	PASS

9. Installation & Execution Guide

Prerequisites

- Python 3.10+ (Standard distribution)
- SQLite3 (Included natively in Python standard library)
- Tkinter & Pillow (**pip install pillow**)

Running the Application

Launch the graphical interface pre-loaded with sample routes, buses, and bookings:

Code Block: POWERSHELL

```
python main.py
```

Running the Unit Test Suite

Execute the automated test suite with verbose output:

Code Block: POWERSHELL

```
python -m unittest test_system.py -v
```

Capturing Automated High-Resolution Screenshots

Run the headless capture suite to regenerate all 9 screenshots into **screenshots/**:

Code Block: POWERSHELL

```
python capture_screenshots.py
```

10. Project Directory Tree

Code Listing

```
Practical/
■■■ models/
■ ■■■ __init__.py # Re-exports all domain entities and database singleton
■ ■■■ bus.py # Bus entity with cabin configuration & priority seats
■ ■■■ database.py # SQLite BusTicketingDatabase singleton with ACID queries
■ ■■■ factory.py # BusFactory & TicketFactory with concession calculations
■ ■■■ passenger.py # Passenger entity with PII & concession ID validation
■ ■■■ route.py # Route entity with travel metrics and pricing
■ ■■■ ticket.py # Ticket entity, TicketStatus enum, & SHA-256 ref generator
■■■ screenshots/ # 9 Programmatic high-resolution PNG screen captures
■ ■■■ 01_tab_booking_seat_map.png
■ ■■■ 02_selected_priority_seat.png
■ ■■■ 03_issued_ticket.png
■ ■■■ 04_tab_manage_bookings.png
■ ■■■ 05_ticket_dossier_inspection.png
■ ■■■ 06_boarding_pass_modal.png
■ ■■■ 07_ticket_cancellation_seat_released.png
■ ■■■ 08_tab_system_analytics.png
■ ■■■ 09_unit_tests_pass.png
■■■ app.py # Main Tkinter GUI application (3 tabs, visual seat map)
■■■ bus_ticketing.db # SQLite persistent database file
■■■ capture_screenshots.py # Automated screenshot generation suite
■■■ main.py # System entry point with sample data pre-population
■■■ test_system.py # Complete 12-test unittest suite
■■■ README.md # Complete project documentation and specifications
```